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System and method for detecting and transmitting incidents of interest of a roadway to a remote server

Abstract

System and methods for automated incident identification and reporting while operating a vehicle on the road using a device. The device identifies incidents using artificial intelligence neural networks trained to detect, classify, segment, and/or extract other information pertaining to objects of interest representing incidents. Additionally, a system and method for further storing, transmitting, processing, organizing and accessing the information graphically with respect to incident type, location, date and time during operation of the vehicle along the road.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Divisional to U.S. patent application Ser. No. 16/930,070, filed on Jul. 15, 2020, all the contents of which are incorporated herein by reference.

FIELD

(1) The present invention is related to image processing for road related incidents.

BACKGROUND

(2) Although many technologies exist for surveying the roads, oftentimes the costs and constraints make it less appealing for practical use. Many solutions require modifications to the vehicle, costly attachments, computers and training to use. Usually these technologies specialize in surveying a single component of road condition. To detect the full spectrum of incidents expected to keep roads maintained, the most common and effective practice is still largely manual. Surveyors drive vehicles down roads and stop their vehicle when they find an object of interest. They take a picture, and then note down the incident details to revisit. This is a slow and costly method, oftentimes leaving a backlog of roads that are overdue for reassessment. As municipalities grow, it becomes harder and harder to stay on top of all the incidents in the system.

(3) Municipalities occasionally outsource road assessment to companies which utilize specialized vehicles employing a wide array of sensors, including cameras, thermal, vibration, radar and laser. While providing a greater level of detail, a specialized vehicle is required to complete the task. The vehicle sensors require extensive calibration, and the processing of the data requires specialized systems and knowledge to produce a report. The costs are extensive, and therefore, such assessments are typically done periodically, where the period duration typically ranges from one year to ten years.

(4) While there have been occasional instances of applications aimed at identifying road damage in video or images, such processes are typically done by first acquiring a data set (in the form of video and geo-positioning data), and then uploading the data to a specialized server where it is processed. The process is cumbersome as often the dataset is too large to transmit over cellular networks. Thus, disadvantageously, current systems must rely upon access to the complete image dataset only once the vehicle has returned to the image processing facility.

SUMMARY

(5) It is an object of the present invention to provide an image acquisition system and/or method to obviate or mitigate at least one of the above presented disadvantages.

(6) Current technologies for surveying roads are extremely costly, requiring complex installation of specialized sensors on a vehicle. They require extensive training and experience to operate and are typically limited only to analysis of pavement condition. Due to the high cost and limited capabilities of the available technologies, municipalities are currently relying on manual inspection to cover the detection of all road related incidents. Further, existing on board imaging systems used for navigation of a vehicle are not optimized for acquiring appropriate images containing objects of interest with respect to a road surface. As such, using a separate device containing a camera that is mounted to the vehicle, in addition to any onboard vehicle imaging systems (used for vehicle navigation), is desired.

(7) Municipalities send their maintenance workers to drive the service vehicles on roads where they must visually locate road related concerns while on patrol or a service call. When workers spot a potential concern they are required to pull over and get off the vehicle to closely investigate the issue and manually initiate an incident or a work order using whatever system they have in place. It may be logged by pen and paper or utilizing a computerized tablet or smart phone.

(8) Getting off the vehicle for each detected issue is not only time consuming but also put workers in a dangerous situation. Contrary to the current technologies available and the method that is implemented to detect road related incidents, the system provided here is capable of automated identification and reporting of a wide variety of road related incidents (e.g. various types of

pavement damage and degradation, road sign damage, road debris, obstructions and other incidents visible from inside of a vehicle).

(9) The system comprising of: device—a mobile computing device such as smartphone, or embedded computer system with built in or peripheral camera(s) and application utilizing neural network(s) and machine learning work together with server(s) that process and store data received from single or multiple devices, serve as gateway to users via web access and present data to users in a meaningful and intuitive manner.

(10) The device is easy to install, can be easily moved to different vehicles and the user does not require training to operate the device. Once the device is started, the device can operate with minimal human intervention.

(11) The user can have a single device mounted on a vehicle or deploy multiple devices deployed on their fleet to have access to an up-to-date insight of their roads. The data created from the incident detection can be used to automate the opening of work orders which helps streamline the process of resolving incidents.

(12) A first aspect provided is a system for capturing digital images containing objects of interest along a first section of a roadway and for transmitting object data pertaining to the objects of interest over a communications network to a server, the server located remotely from the system, the system comprising: a device having: a camera for obtaining the digital images; at least one sensor for acquiring sensor data pertaining to the objects of interest contained in the digital images; and a memory and processor for executing image processing instructions for processing the digital images in order to generate the object data, the object data including the sensor information and the resultant data representing the objects of interest; and a mounting component for mounting the device to a vehicle, the vehicle for travelling along the roadway during operation of the camera; and a network interface for sending the object data over the communications network to the server during operation of the vehicle on the roadway.

(13) A further aspect provided is a method for capturing digital images containing objects of interest along a first section of a roadway and for transmitting object data pertaining to the objects of interest over a communications network to a server, the server located remotely from the system, the method stored instructions for execution by a processor: obtaining the digital images by a camera mounted by a mounting component to a vehicle, the vehicle for travelling along the roadway during operation of the camera; acquiring sensor data pertaining to the objects of interest contained in the digital images; executing image processing instructions for processing the digital images in order to generate the object data, the object data including the sensor information and the resultant data representing the objects of interest; and sending the object data over the communications network to the server during operation of the vehicle on the roadway.

(14) A further aspect provided is a system for collecting, processing, discarding, storing, transmitting, and disseminating data, the data includes information captured from one or more cameras and one or more sensors mounted in a vehicle, and generated object data and generated resultant data, the system comprising: at least one device per vehicle, the at least one device having: a housing; one or more processors for executing software instructions; one or more memory components; one or more network interfaces; the one or more sensors for acquiring sensor data; the one or more cameras for obtaining digital images; a mounting component for mounting the housing of the at least one device to the vehicle; and the software instructions, including image processing operations and related software instructions to discard image data, generate the object data and the resultant data; and one or more servers, communicating with the at least one device over a communications network, the at least one device mounted in one or more vehicles.

(15) A further aspect provided is a method for collecting, processing, discarding, storing, transmitting, and disseminating digital data, the data includes information captured from a device, a camera and a sensor mounted in a vehicle, and generated object data and generated resultant data.

(16) A further aspect provided is a system for collecting, processing, discarding, storing,

transmitting, and disseminating data, said data includes data captured from camera(s) and sensor(s) mounted in a vehicle, and generated object data and resultant data, the system comprising of: one or more devices per vehicle, having: a housing; one or more processors for executing software instructions; one or more memory components; one or more network interfaces; one or more sensors for acquiring sensor data; one or more cameras for obtaining digital images; a mounting component for mounting the housing of the device to a vehicle; and software instructions, including image processing operations and related software instructions to discard image data, generate object data and resultant data; and one or more servers, communicating with one or more devices, mounted in one or more vehicles.
(17) A further aspect provided is a method for collecting, processing, discarding, storing, transmitting, and disseminating digital data, said data includes data captured from device, camera(s) and sensor(s) mounted in a vehicle, and generated object data and resultant data.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Exemplary embodiments of the invention will now be described in conjunction with the following drawings, by way of example only, in which:
(2) FIG. **1** shows components of a system for automatically detecting objects and incidents, and storing and transferring them to a server;
(3) FIG. **2** shows an example series of images of the system of FIG. **1**;
(4) FIG. **3** shows options for powering a device of the system of FIG. **1**;
(5) FIG. **4** shows an example split processing framework for the system of FIG. **1**;
(6) FIG. **5** shows device camera options and examples of objects that different lenses are used to capture of the system of FIG. **1**;
(7) FIG. **6** shows an example of image capture, redaction and other processing before being encrypted and stored in device or transmitted to the server of the system of FIG. **1**;
(8) FIG. **7** shows example sensors used by the device to detect and store additional information along with each of the incident images captured of the system of FIG. **1**;
(9) FIG. **8** shows example image processing operations of the system of FIG. **1**;
(10) FIG. **9** shows example workflows of image processing functions and neural networks processing of the system of FIG. **1**;
(11) FIG. **10** illustrates different ways of identifying objects of interest using different neural networks of the system of FIG. **1**;
(12) FIG. **11** shows different kinds of information on a display of the device of the system of FIG. **1**; and
(13) FIG. **12** shows automated incident identification reporting to users of the system of FIG. **1**.
DESCRIPTION
(14) Referring to FIG. **1**, shown is a system **10** designed to automate identification and reporting of road related incidents/conditions/objects **12** (e.g. pavement damage, street sign damage, debris on road, etc.) with respect to a road surface **14** and/or adjacent road surface **14** surroundings **13** in real time. The road related incidents/conditions/objects **12** are hereafter referred to generically as objects **12**, for ease of description purposes only. Digital images **16** are taken by one or more imaging devices **101**, e.g. a smartphone or other portable imaging device preferably with network connection capabilities to a communications network **18** (e.g. wireless network **105**, cellular network **106**, etc.). Only selected data portions **20** (e.g. image frames **16***a*,*b*, *c*, *d*—see FIG. **2**) of the images **16** can be transmitted over the network **18** by the device **101** to a server **107** for subsequent processing/reporting, as further described below, such that image discard data **19** is excluded (see FIG. **4**) from object data **21** transmitted to the server **107**. Alternatively, unprocessed

image(s) **16** can be transmitted by the device **101** over the network **18**, as they are acquired in real time (subject to network **18** connectivity constraints), to the server **107** for subsequent processing. It is also recognized that both resultant processed image data **20** and unprocessed images **16** can be transmitted to the server **107** by the device **101** over the network **18** (e.g. as object data **21** including the sensor information **17**). In any event, it is recognized that it is advantageous for the system **10** to be configured to preferably send object data **21** by the device **101** to the server **107** as the object data **21** is acquired, in order to take advantage of image **16** processing and acquisition efficiencies as discussed.

(15) It is recognized that one advantage of the current system **10** is that storing and/or sending of the object data **21** can occur in real time while the device **101** continues collection and processing of additional digital images **16** for a second section of the roadway **14** different than the first section of the roadway **14** associated with the sent object data **21**. As such, the system **10** is configured to continuously store/send respective sets of object data **21** for respective differing sections of the roadway **14** on a real time basis and/or scheduled basis, dependent upon appropriate network **18** connectivity between the device **101** and the server **107**. Accordingly, sending of the object data **21** over the network **18** can include queuing a set of object data **21** in memory **99**, **104** for before transmission to the server **107**.

(16) The device **101** is mounted by a mounting component **103** to a body **22** of a vehicle **102** (e.g. car, truck, bus, etc.), such that an imager **500** (e.g. camera **500**—see FIG. **5**) has a viewpoint of the road surface **14** and optionally any desired adjacent surroundings **13** (e.g. roadside, overhead, etc.). Accordingly, during operation of one or more vehicles **102** along the road surface **14**, the imager(s) **500** of the device record/capture the images **16** according to the configured viewpoint (as facilitated by orienting the device **101** with respect to the road surface **14** and/or adjacent surroundings **13** as per the capabilities of the mounting component **103**). It is recognized that the captured images **16** will contain recorded data of the objects **12**, which will be identified by image processing instructions **905** (stored in memory **99** provided by computing infrastructure **100** of the device **101** —see FIG. **7**), also referred to generically as neural network(s) **905**.

(17) Referring to FIGS. **1** and **7**, the system **10**, as an example, can utilize device **101** (e.g. a mobile computing device such as smartphone, smart camera or embedded computer system) physically coupled to the vehicle **102** (an automobile to which the device **101** is attached) by the mounting component **103** (i.e. attaching the device **101** to the vehicle **102**). In general, the device **101** executes software **108** (i.e. stored instructions) which collects image data **16** (the digital images **16**) from the device's **101** camera(s) **500** (the imagers **500**) and infers (i.e. determines) any objects **12** present in the images **16** using the neural network(s) **905** (e.g. image processing instructions **905**) by way of a computer processor **111** and/or a graphics processor **112**. The device **101** transmits the data **20** (containing information about the objects **12** identified from the images **16**) along with acquired sensor **700** information **17** to the server **107** (via the network **18** utilizing the cellular connection **106** or wireless connection **105**, for example). The device **101** can store data (e.g. images **16**, processed data **20**, sensor information **17**) in non-volatile memory **104** (of the computing infrastructure **100**) prior to transmitting any of the data **20**/sensor information **17** over the network **18**. The processed data portions **20** (containing the identified object(s) **12** of interest) and/or sensor information **17** can be referred to generically as object data **21** (see FIG. **1**). As discussed herein, the embodiment of image **16** processing by the device **101** is considered only one example. As such, it is recognized that processing of images **16** can be performed by the device **101** alone, by the server **107** alone, or by both the server **107** and the device **101** as provided for in FIG. **4** for the shared/distributed processing environment **400**.

(18) The server **107** can facilitate as gateway to users (of the device **101**) to make data available to other users (e.g. road supervisors, construction managers, etc.) in a meaningful and intuitive manner by way of accessing any of the data **20**/sensor information **17** received by the server **107**. It is recognized that as one embodiment, the resultant processed data **20** containing information of the

identified objects **12** (from the images **16**) can be portions of the images **16** themselves. Alternatively, or in addition to, the resultant processed data **20** can also include metadata (e.g. descriptions/descriptors) of the objects **12** identified from the images **16** by the image processing instructions **905**. For example, the descriptions/descriptors of the objects **12** can include object type (e.g. road sign, pothole, road debris, etc.), object size (e.g. 10 cm wide by 20 cm deep), etc. In any event, it is recognized that the resultant processed data **20** represents only a portion of the total images **16** data recorded by the imager **500**, such that images **16** (and/or the descriptions/descriptors representing the images and their image contents) not containing objects **12** (of interest) are excluded (e.g. the image discard data **19**) from the resultant processed data **20** sent over the server **107** over the network **18**, as one example embodiment.

(19) Device **101**

(20) It is recognized that the device **101** is an integral part of the system **10**. Referring to FIGS. **1** and **7**, the device **101** can be consist of a plurality of hardware and software components that are configured to automatically collect visual, location and sensor data (e.g. images **16** and sensor information **17**) while affixed to the vehicle **102** travelling along the road surface **14**. Components that typically make up the device **101** are embedded in the computing system/infrastructure **100**, typically inclusive of a central processing unit (cpu) **111** and/or a graphics processing unit (gpu) **112** and memory **99**, **104** including a high speed volatile memory such as a ram as well as non-volatile memory, any of which facilitates the device **101** to execute its software **108** (e.g. operating system, image processing instructions **905**, etc.). The device **101** may also have a read only memory **99**, **104** for storing instructions necessary for operation of the device **101**. Further, the non-volatile memory **104** can be associated with storing files **706** associated with operating system(s), component driver(s), application(s), and media, alongside other software applications resident in memory **99** of the device **101**. The device operating system (e.g. part of the software **108**) can be such as a windows operating system, android operating system, linux operating system, or other operating system, whether embedded or not.

(21) The device **101** can also have one or more data transmitting and receiving components (communication components operating a network interface **113** to the network **18**) which can be wireless. Further, the device **101** can interface with external wireless communication components **117** via wired connector **115**, such as a usb connection or an ethernet connection, in order to transmit the data **20** and sensor information **17** over the network **18** to the server **107**, as an example.

(22) In terms of sensors **700**, the device **101** can have, by way of example, a geo-location sensor **701** or a geo-positioning sensor **701**, provide location based information using satellite (such as gps, glonass, galileo) or cellular tower locations to determine device positioning information (as part of the sensor information **17**). The device **101** may also utilize sensors **700** related to its positioning, location and orientation such as accelerometer sensor **702**, gravity sensor/gyroscope sensor **703**, rotational axis sensor **704** and/or other sensors **705**. The device **101** will typically utilize a battery **307** (see FIG. **3**) which can facilitate operation of the device **101** even when connection to vehicle power **302** is temporarily interrupted. For example, in the event that the vehicle **102** is temporarily shut-off, such as when filling gas, the device **101** can continue to operate on its battery **307** charge, in order to effect the image processing of the acquired images **16** and subsequent sending of the resultant processed data **20** and sensor information **17** over the network **18** (see FIG. **4**), as an example.

(23) Further, the device **101** can have a user interface **119** including a display, built-in or external, in order to display information from the device **101**, to an operator of the device **101**, the display information such as but not limited to the camera **500** field of view (viewfinder), the orientation of the device **101**, status indicators, settings, parameters, and other information related the installation, configuration operation, and maintenance of the device **101**.

(24) As further described below, the device **101** can execute the software **108** (including the

artificial intelligence neural network(s) **905**) for detection and classification of objects **12** of interest in the acquired images **16** in order to infer (e.g. determine) what object(s) **12** are present in images **16** and/or the position of the object **12** in the images **16**. The software **108** can also collect, process, store and transmit sensor information **17** from its sensors **700** for the geo physical location (e.g. GC a, b, c, d—see FIG. **2**) in which the image **16** was acquired with respect to the road surface **14** (i.e. geographical coordinates of the portion of the road surface **14** that contains the objects(s) **12**).

(25) Further to the above, it is recognized that the device **101** components can be packaged together in a common housing (not shown) or some components can reside externally to the device **101** and connected via currently common interface connector **115** such as a USB port. Examples of electronic devices that can be configured to function as the device **101** can include smart phones, smart cameras and embedded computer systems. For example, a smartphone **101** can be defined as a mobile device that contains components within to run the software **108**. This device **101** can be ideal for a portable installation facilitating for easy transfer between different vehicles **102**. Example of currently available capable smartphones are Samsung galaxy s10, s10+, s20 models, Samsung note s10, s10+ models, LG g8, iPhone 11 pro, iPhone 11. It is expected that many of the newer smartphone models by the majority of the smart phone manufacturers can also perform as a device **101** when enabled with the software **108** (i.e. including the neural network(s) **905** as further described below).

(26) For example, a smart-camera **101** can be defined as a camera with image processing capabilities (e.g. software **108**) that is capable to execute the instructions of the artificial intelligence neural network(s) **905**. For instance, a camera **101** containing artificial intelligence enabled chipsets such as Intel Movidius, Nvidia Cuda, Texas Instrument Sitara, Qualcomm Adreno series, etc. Alternatively, the camera **101** can be packaged together with an embedded cpu **111** (such as the likes made by companies such as Intel, Amd or Arm) which facilitates execution of the artificial intelligence neural network **905**. The camera sensors **700** and transmission components (e.g. network interface **113**) can be embedded or externally connected, as desired.

(27) For example, the embedded computer system **101** can be defined as a computing device **101** that is designed to operate (via the software **108**) with some resistance to shock, vibration and temperature fluctuations. Sensors **700** are embedded and/or provided as peripheral (i.e. External) devices. The embedded computer system **101** can be considered for permanent installation in the vehicle **102** or for installations with multiple cameras **500**. Current examples of embedded computer systems **101** include Nvidia Jetson AI platform, Google coral edge series, raspberry pi series, rugged industrialized embedded computers specifically environmentally hardened for use vehicles **102**, or other computing devices **101**.

Example Embodiments for Imagers **500**, Also Referred to as Cameras **500**

(28) Referring to FIG. **5**, the device **101** can be equipped with internal camera(s) **501** or external camera(s) **502** which can be of different types such as telephoto **503**, wide angle **504**, or night vision **505**, or other types of cameras **500** (e.g. imagers **500**), used to record a digital image **16** as acquired by the device **101** as the vehicle **102** traverses along the road surface **14**. It is recognized that some of the images **16** acquired will contain object(s) **12** of interest, while others of the acquired images **16** will not contain any desired object(s) **12** of interest. As further discussed below, the acquired images **16** can be embodied as a series/plurality of image frames **16***a*, **16***b*, **16***c*, **16***d*, for example (see FIG. **2**). The image frames **16***a*, **16***b*, **16***c*, **16***d* comprise the plurality of images **16** acquired by the device **101** as the vehicle **102** travels along the road surface **14** (see FIG. **1**).

(29) As the device **101** primary sensor is a camera **500**, the device **101** can include at least one camera **500**. Depending on the configuration of the device **101**, the device **101** can encase one or more cameras **501** (for example, if the device **101** is a modern smartphone). The device **101** can also be encased in a camera **501** (for example, if the device **101** is a smart camera, or an ai enabled camera). The camera(s) **500** can also be attached to the device **101** externally **502**. For example, the

device **101** can be an embedded computer connected to an external camera via a wired or wireless interface. The device **101** can utilize internal camera(s) **501** and external camera(s) **502** at the same time. For example, a smart phone **101** can utilize its built in camera **500** to process images **16** acquired facing the front of the vehicle **102**, whereas wireless cameras **500** could be mounted on the sides of the vehicle **102** transmitting images **16** wirelessly to the smartphone **101**.

(30) The device **101** can have different type(s) of camera(s) **500**. Some examples of different types of cameras **500** include telephoto **503**, whereas the camera **500** is optimized to capture images in the distance; wide angle **504** camera(s), whereas the camera **500** is optimized to detect an image **16** with a wide field of view; and/or a night vision **505** camera, whereas the camera **500** is optimized to acquire images at low light settings.

(31) Different camera(s) **500** can be used for different use cases. For example, a wide angle camera **504** may be used to detect issues with signage **12** (for example, a damaged sign), whereas a telephoto **503** camera can be used to detect road defects **12** (for example, cracks or potholes); and an infrared camera **505** can be used for night time image **16** acquisition. The types of incidents **12** noted in FIG. **5** are for illustration and do not mean that the incidents **12** depicted are exclusive to the camera **500** type as given by example.

(32) Further, a device **101** can utilize multiple cameras **500**, internal and/or external simultaneously. For example, the device **101** can be attached to a vehicle's **102** windshield and have both a telephoto **503** and a wide angle **504** camera internally **501**. The telephoto **503** camera may be used to detect road defects **12** (for example, cracks and potholes) and road issues **12** (for example, faded lane markings or open manholes), while at the same time, the wide angle **504** camera can be used to detect damaged signs **12**. The same device **101** can also be connected to two external **502** cameras **500** mounted on the vehicle's **102** side windows, facing the roads and curb as part of the desired road surface **14** and surroundings **13** in the camera's field of view. The external cameras **500** can look for road damage **12** and curb damage **12**.

(33) In some use cases, different cameras **500** can be used under different circumstances. For example, at night when lighting conditions are poor, the device **101** can switch from one camera **500** to another optimized for night vision **505**. This may be done programmatically with the use of the software **108**, based on schedule (for example, based on a sunset timer), image parameters (such as exposure or brightness), or a sensor **700** (for example, when a light sensor determines it is less than certain luminance level). It can also be done manually by a driver selecting a different camera **500** or selecting a "night mode" which includes the night camera **505**. Similarly, in some use cases, the same cameras **500** can be used with different camera settings. Examples of settings that may be adjusted for night time operation include frame rate, resolution, aperture, ISO, brightness, and/or shutter speed. As an example, at night when lighting conditions are poor, the device **101** can switch from daytime camera settings to nighttime camera settings. This may be done programmatically with the use of software **108**, based on schedule (for example, based on a sunset timer), image parameters (such as exposure or brightness), or a sensor (for example, when a light sensor determines it is less than certain luminance level). It may also be done manually by a driver selecting a different camera or selecting a "night mode" which includes the camera night time settings.

(34) Some digital cameras **500** can give access to almost all the image **16** data captured by the camera **500**, using a raw image format. An example type of the cameras/imagers **500** can include digital image sensors using metal-oxide-semiconductor (MOS) technology. Another type is digital semiconductor image sensors, including the charge-coupled device (CCD) and the CMOS sensor. Another type can be the NMOS active-pixel sensor (APS).

(35) Images **16**

(36) The vehicle **102** uses imagers/cameras **500** to record the series of digital images **16** (as acquired by the device **101**) while the vehicle **102** traverses along the road surface **14**. It is recognized that some of the images **16** acquired will contain object(s) **12** of interest, while others of

the acquired images **16** will not contain any desired object(s) **12** of interest, as determined by the image processing instructions **905**. As further discussed below, the acquired images **16** can be embodied as a series/plurality of image frames **16***a*, **16***b*, **16***c*, **16***d*, for example (see FIG. **2**). The image frames **16***a*, **16***b*, **16***c*,**16***d* comprise the plurality of images **16** acquired by the device **101** as the vehicle **102** travels along the road surface **14** (see FIG. **1**). The images **16** can be embodied as captured individual still image frames **16***a*, **16***b*, **16***c*, **16***d* and/or embodied as captured video containing the individual frames **16***a*, **16***b*, **16***c*, **16***d*, as desired. As shown by example, image frames **16***b* and **16***d* contain objects **12**, while image frames **16***a* and **16***c* do not contain any objects **12**. Further, it is recognized that the plurality of image frames **16***a*, **16***b*, **16***c*, **16***d* are captured sequentially with respect to one another, i.e. distributed sequentially over recorded time and distance along the road surface **14**, as further discussed below. As discussed, the images **16** may actually contain potential objects **12** of interest, (e.g. a pothole), however these potential objects **12** of interest could be processed as discarded image data **19** and thus not included in the resultant processed image data **20** (e.g. as provided by the device **101** and/or the server **107**).

(37) One example of discarding a potential object **12** is where a geo coordinate GC a, b, c, d matches an already transmitted object **12** of interest in a previous transmission object data **21** to the server **107**. In this manner, duplication of objects **12** in the data **20** can be advantageously inhibited. Another example where a potential object **12** is retained in the resultant processed object data **20** is where the geo coordinate GC a, b, c, d does match the GC a, b, c, d of the object **12** in a previous transmission of object data **21**, however a state of the potential object **12** has changed (e.g. a state such as a size of the object—a pothole size of the potential object **12** has increased over the size of the same object **12** reported in previously transmitted object data **21** for the same pothole **12** at that identified GC a, b, c, d).

(38) For example, as shown in FIG. **2**, the selected image frames **16***a*, **16***b*, **16***c*, **16***d* of the plurality of images **16** provide for some imaged portions **14***a,b,c,d* of the road surface **14** or surroundings **13** containing objects **12** while others do not. As an illustrative example, the respective geo coordinates GCa, b, c, d (e.g. sensor information **17***a,b,c,d*) for each image frame **16***a,b,c,d* of the images **16** can be used to represent contiguous sections (overlapping or non overlapping in geographical coordinates) of the road surface **14** or surroundings **13** portions **14***a,b,c,d*. As such, it is clear from the example image frames **16***a,b,c,d* provided that image road surface/surrounding **14***a* (in image frame **16***a*) and road surface/surrounding **14***c* (in image frame **16***c*) do not contain any objects **12**. On the contrary, image portion **14***b* (in image frame **16***b*) and image portion **14***d* (in image frame **16***d*) do contain objects **12**. Further, for example, geo coordinates GCa, b, c, d can represent physical geographical coordinates (e.g. provided by sensors **700** of the device **101** associated with at what geographical position the vehicle **102** was at on the road surface **14** when the particular image frame **16***a,b,c,d* was recorded). As such, as further discussed below, the image processing instructions **905** are used by the device **101** to determine that image frames **16***b,d* contain object(s) **12** and image frames **14***a,c* do not contain object(s) **12**. Further, it is recognized that each image frame **16***a,b,c,d* can have a positional reference frame REFa, b, c, d, such that the physical location of the object **12** within the image frame **16***b,d* can be specified. Also as discussed below, it is a distinct advantage of the current system **10** to have an ability to identify those images **16** (e.g. image frames **16***a,c*) that do not contain any objects **12** of interest and thus can be discarded and thus not be included as part of the object data **21** (e.g. transmitted to the server **107**—see FIG. **1**). As further discussed below, it is recognized that just because a particular image frame **16***a,b,c,d* contains an object **12**, as identified by the image processing instructions **905**, the identified object **12** may still subsequently be discarded by the image processing instructions **905** for a number of reasons (see below).

(39) As such, any discarded image data (e.g. discarded data **19**—see FIG. **4**)) from the images **16** constitutes bandwidth transmission savings for communication of the object data **21** between the device **101** and the server **107**. Further, any discarded image data **19** (e.g. entire frames **16***a,b,c,d*

and/or selected portions of an image frame **16***a,b,c,d*) from the transmitted object data **21** can also result in downstream data processing/storage savings by the server **107**. It is clear, as discussed below, that the system **10** can take advantage of a split image processing framework **400**, see FIG. **4** further discussed below, such that a first portion **402** of the image **16** processing can be performed on the device **101** itself (i.e. prior to transmission of the object data **21** to the server **107**), while a second portion **404** of the image **16** processing can be performed on the server **107** post receipt of the object data **21**. For example, the image processing instructions **905** can be used as a first image processing portion **402** to identify objects **16** of interest and thus discard/exclude data **19** representing entire frames **16***a,b,c,d* (or portions of frames **16***a,b,c,d*) before transmitting the object data **21**. The server **107** can then be tasked with identifying the relevance of the objects **12** of interest in the object data **21**, including for example the physical position (e.g. REFa, b, c, d) of the objects **12** in the frames **16***a,b,c,d* included in the object data **21** as part of the second image processing portion **404**. It may also discard or re-acquire/refresh incident images **16** based on a timer that expires (for example, a certain number of days has elapsed) or based on distance from previously noted incident (as the GPS may not be 100% precise), and as such, data **19** can be discarded based on previously recorded incident.

(40) Accordingly, the image discard data **19** is considered as those portion(s) of image **16** data that does/do not contain determined object(s) **12** of interest by the image processing instructions **905**, e.g. as implemented by the processor **111** of the device **101**. Examples of the image discard data **19** are shown in FIG. **6**. For example, image information **601**, **604**, **605** shows example image data content **20** as included (e.g. image data of the images **16** representing the identified object(s) **12** of interest) as well as image discard content **19** as discarded. As discussed, the discarded image content **19** can be withheld from inclusion in the object data **21** transmitted to the server **107** (see FIGS. **1** and **4**).

(41) In terms of format, a digital image **16** can be an image **16** containing digital data content (e.g. both data **19** and data **20** as present prior to processing) representing picture elements, also known as pixels, each with finite, discrete quantities of numeric representation for its color intensity or gray level that is an output from its two-dimensional functions fed as input by its spatial coordinates denoted with x, y on the x-axis and y-axis, respectively (e.g. positional reference frame REFa, b, c, d). The image would be acquired either as raw image data available in various formats such as YUV RGB, HSL, HSV, or other image color spaces and encodings as available from the camera **500** device. The data (e.g. both data **19** and data **20** as present after to processing) may be available in the form of coordinates which can be scaled similar to vector images. Vector images can have the unique advantage over raster graphics in that the points, lines, and curves may be scaled up or down to any resolution with no aliasing. The points determine the direction of the vector path; each path may have various properties including values for stroke color, shape, curve, thickness, and fill. As such, it is recognized that part of the image **16** acquisition and subsequent image processing (by the image processing instructions **905**) can be used to incorporate data **19**, **20** which can be overlayed on the image in a vector graphics or a separate raster image and flattened/merged into the image **17** itself to display the detection **12** in the image **16**. The data **21** may be overlayed/incorporated into the image data **20** or stored and sent separately, with association to the respective image **16**. As discussed above, images **16** can be acquired through digital cameras **500**, which use any of several image file and color space formats known in the art.

(42) It is recognized that the images **16** can be compressed using any known compression technology, before they are sent as part of the object data **21** (as discussed) to the server **107**. It is also recognized that image compression is not used by the device **101** (e.g. by the system **900**—see FIG. **9**) to identify objects **12** of interest in the images **16**, in order to produce the processed image data **20**. In other words, data compression can be utilized downstream of generation of the processed image data **20** (used to separate or otherwise remove the image discard data **19** from the overall plurality of image data **16** acquired initially by the camera(s) **500** of the device **101**).

(43) For example, the camera **500** and/or the image processing instructions **905** can utilize digital image compression, for those portions of the images (e.g. frames **16***a,b,c,d* determined to contain object(s) **12** of interest), before transmitting such image data **20** in the object data **21** (e.g. see FIG. **1**). For example, digital image compression technology can incorporate discrete cosine transform (DCT), a lossy compression technique. For example, as an image format for any image data of the object data **21**, JPEG compresses the image content down to relatively smaller file sizes. For example, the DCT compression algorithm of the JPG format can be used.
(44) Vehicle **102**
(45) The device **101** is intended to be used when deployed in the vehicle **102**. In many cases, the vehicle **102** can be a vehicle that is operated on behalf of an organization which can be governmental, quasi-governmental or a private company. It can also be used voluntarily by individuals as a crowd-sourced application.
(46) Examples of governmental organizations include all levels of government, including national, federal or republic government; provincial, territorial or state government; municipal government, including municipalities, upper tier municipalities (such as metropolitan, regional, county or other name used to describe major upper tier municipalities), or lower tier municipalities (such as city, village, township, town, community, or other name used to describe). The governmental organization may also be a special organization, such as a reserve, resort, other name that is used to describe the local government of a certain geography and population. Examples of quasi-governmental organizations would be government-owned or supported organizations. Those could be organizations established as part of a public-private-partnership or a concession to build, maintain and/or operate an asset or a service over a period of time. They could be separately incorporated but the government may have full ownership, majority ownership, or minority ownership. The government's representatives can sit on the board of such organizations. Examples of quasi-governmental organizations include toll road concession companies, bridge concession companies, transportation and/or transit authorities, and/or utility or telecom companies. A private company can simply be a private company that is the owner of the asset that is to be inspected, or contracted on behalf of the owner to do so.
(47) The vehicle **102** can be a service vehicle dedicated to patrolling an area for the specific purpose of identifying incidents/objects **12** on behalf of the organization. The vehicle **102** can be a car, a truck, a golf cart, a tractor, an atv, a bicycle, an e-bike, a motorbike, a motorcycle, a snowmobile, a van, or a customized utility vehicle, for example. The vehicle's **102** primary purpose can be different than acquiring incidents/objects **12**, but augmented with the device **101** mounted for a supplemental function of incident/object **12** detection on behalf of the organization. Examples include garbage trucks, snowplows, operational service vehicles, utility vehicles, road and sidewalk sweepers, public transportation vehicles such as buses, school buses, and transportation vans or taxis. The vehicle **102** can also be a private vehicle owned by an individual, whereby the individual contributes incident object **12** data detected by the device **101** on a good will to the organization, or for monetary compensation. The vehicle **102** can also be an autonomous vehicle **102** operated privately or by an organization, whereas the autonomous vehicle **102** can be equipped with the device **101** to automatically detect incidents/objects **12** in the area in which it operates.
(48) Mounting Component **103**
(49) The device **101** is intended to be mounted in the vehicle **102** using the mounting component **103**. Typically, the device **101** can be mounted to the vehicle's **102** windshield (e.g. a portion of the body **22**—see FIG. **1**), though it can also be attached to the dashboard, the side windows, the back windows, or the vehicle **102** frame, as desired. The mounting component **103** configuration can be different for different vehicle **102**/device **101** combinations. For example, depending on whether the device **101** is a smartphone, a smart camera, or an embedded computer with external camera different mounting configurations can be used. Different mounting configurations can also be used depending on whether the device **101** is to be permanently affixed to the vehicle **102** or transferable

between different vehicles **102**. The vehicle **102** can utilize the mounting component **103** of different types. For example, the mounting component **103** can be attached to the vehicle **102** via a suction cup, a sticky tape, glue, or screws and bolts, for example. The mounting component **103** can allow for an easy removal of the device **101** by having the device **101** easily detach from the mounting component **103**. The mounting component **103** itself can also have multiple parts which facilitate detaching parts of the mounting component **103** together with the device **101**.

(50) Non Volatile Memory **104**

(51) The storage capabilities (e.g. memory **99**, **104**, etc.) of the device **101** can have the non-volatile memory **104** associated for storing files **706** associated with operating system(s), component driver(s), application(s), and media, alongside other files used in software applications. The non-volatile memory **104** can be embedded in the device **101**, an add-on, and/or a peripheral. For example, smartphones currently come with built in non-volatile memory **104**, which can be expanded using non-volatile memory **104** add-ons (for example, microsd memory). Embedded computers typically come with a variety of hard drives, flash drives, and interfaces which allow for including one or more non-volatile memory **104** storage components.

(52) The device's **101** software **108** can store digital data, such as the acquired incident images **16**, data portions **20** of the images **16** including data about the identified objects **12** from the images **16**, and associated sensor **700** data (e.g. sensor information **17**) onto the non-volatile memory **104**. It is recognized that the data portions **20** are those determined excepts (e.g. frames **16***a,b,c,d*) from the images **16** that contain the objects **12** of interest.

(53) The device **101** (e.g. via software **108**) can store the images/data **21** to the non-volatile memory **104** prior to transmitting over the network **18** for a variety of reasons. For example, cellular **106** connectivity may not be reliable, and the data **21** would have to be stored temporarily before transmitted. If cellular connection **106** or wireless connection **105** to the internet **18** is not available (e.g. out of cellular network coverage or out of the wireless range) during operation, the software **108** will store the images **16**, **20** and their associated data **17** to the device's **101** non volatile memory **104**. Once the connectivity has been restored, the software **108** will resume uploading the object data **21** to the server **107**.

(54) Some organizations may also opt to utilize wireless connectivity **105** in order to save on cellular data costs, and as such, the data **21** would be stored until such time that the device **101** has access to wireless **105** access point, which is connected to the internet **18**. The device **101** may also benefit from a performance, power, and/or heat management perspective to only transmit data once the vehicle **102** is idle (for example, idle in traffic or in a parking lot). The device **101** may also be benefit from a performance, power, and/or heat manage perspective to initiate an upload process as a scheduled process as opposed to an ongoing process.

(55) The files **706** and/or database **707** (representing any of the images **16**, sensor information **17**, and/or data portions **20**—i.e. processed images **16**) may be in an encrypted format in the memory **104** to secure the information. The files **706** and/or database **707** are deleted from the non volatile memory **104** when they are successfully uploaded to the sever **107**. The data **16**, **17**, **20** can be stored on the non volatile memory **104** in various file formats. Image data can be stored as compressed or uncompressed files such as jpegs, bitmaps, webp, png, and other common image formats. The associated sensor data **17** can be stored as a metadata file (for example, an xml file), a table file (i.e. a csv or a txt file), or in the database **707**. The files **706** may also be stored in a common file **706** format or in a proprietary one.

(56) Wireless Connection **105**

(57) The device **101** can include the wireless connection **105** network interface in order to connect to the server **107**. For example, a wireless connection **105** can be may be wireless lan or wi-fi, operating at common frequencies such as 2.4 ghz, 5 ghz, or other frequencies typically associated with ieee 802.11 or other wireless standards. The wireless connection **105** has the advantage that it typically allows to upload a large volume of data without the cost of cellular data usage.

(58) Cellular Connection **106**

(59) The device **101** can include a cellular connection **106** network interface in order to connect to the server **107**. For example, the cellular connection **106** can utilize technologies such as 3g, 4g, lte, 5g or other technologies used for access to cellular towers. The cellular connection **106** provides for communications with the server **107** on a constant, frequent or periodic basis, allowing the incidents **12** (contained in the object data **21**) to be generated as communications are taking place. It is expected to be used in many scenarios including when cellular cost is not a major issue, when cellular connection is available, when faster response to incidents is necessary, and/or when the device **101** does not have access to wireless connectivity **105**.

(60) Server **107**

(61) The server **107** is responsible for the organizing, storing, processing and disseminating of the object data **21** uploaded by the device(s) **101**. A single server **107** can host a plurality of users, whether governmental users, quasi-governmental users, or private organizations. A single server **107** can communicate with a plurality of devices **101** as clients of the server **107**. The object data **21** of each client is securely segregated from each other where one user cannot access the object data **21** of other users, unless a user purposely marked their data for sharing, for example. When the server **107** receives object data **21** from a device **101** through the internet **18**, the image data (e.g. frames **16***a,b,c,d*) containing objects **12** of interests are stored in a server storage **30** to a folder allocated specifically to the particular device **101** and user's organization. It can be further organized by date, road segments, or other hierarchal structure, as desired. The server **107** can store image data **16**, sensor data **17**, object data **21**, resultant processed data **20**, resultant processed data **20′** and/or discard data **19**, **19′**, as desired. For example, the device **101** can send the discard data **19** to the server **107**, as part of the object data **21** or subsequent to sending the object data **21**, as desired.

(62) Accordingly the discard data **19**, **19′** can be defined as the data removed from the images **16** in order to produce the object data **21** (e.g. containing the processed images **20**). For example, the discard data **19** could be inclusive of one or more whole frames/images **16***a,b,c,d* that are dropped (e.g. removed from the images **16** and/or sensor data **17**) before sending the resultant object data **21** (as a result of utilizing the image processing instructions **905** on the captured images **16** and/or sensor data **17**). For example, the image processing instructions **905** can also include utilizing the sensor data **17** in order to identify the objects **12** and/or discard data **19**. For example, the sensor data **17** (such as direction and/or GPS data) can be used by the device **101** to recognize and extract the discard data **19** from the image data **16**. For example, images **16** captured are associated with sensor data **17** during their capture (e.g. direction of vehicle travel and GPS data for recognizing the geo position of the captured images **16**). As such, when new subsequent images **16** are taken, having sensor data **17** (e.g. direction and/or GPS data) matching that sensor data **17** of the previously recorded images **16**, the newly acquired images **16** can be discarded (either in whole or in part). The newly acquired images **16** could be considered by the device **101** (as a result of executing the image processing instructions **905** on the previous images **16** and newly acquired images **16** and their associated sensor data **17**) as duplicate data in view of the match in sensor data **17** between the previous images **16** and the newly acquired images **16**.

(63) Additional information, such as sensor data **17**, geographical coordinates GCa, b, c, d, direction of travel, date/time, pitch, and/or data of the object data **21** describing the object **12** of interest in the image can be associated to each incident image (e.g. image frame **16***a,b,c,d*) and also stored into one or more database(s) **30** on the server **107**. The server **107** is depicted in the system **10** as a single server **107**. While it may be deployed as a physical server, there may be more than one server **107** segmented by geography (for example, Canada, Us, Mexico or other countries), by architectural function (DNS, runtime, database, storage, image processing, reporting, or other function), by capacity (for example, users 1-1000 reside on server 1, whereas users 1001-200 reside on server 2), logically (such as a virtual machine that runs on a server cluster, or on a cloud), or in

other common ways in which servers which provide software as a service are setup. For greater clarity, the word server **107** and server(s) **107** will be used interchangeably throughout the description, figures and claims as the system **10** could be setup to use one or more physical, virtual or cloud based servers **107**.

(64) Software **108**

(65) The software **108** can be responsible for acquiring image **16** data from the camera **500**, acquiring geographical positioning data **17** from the geo-positioning sensor **701**, collecting sensor(s) data **17**, collecting other system information such as date and time, identifying incidents **12** using image processing function(s) and neural network(s) inference workflow(s) of the image processing instructions **905**, and reporting the identified incident(s) **12** (considered of interest) through communications of the object data **21** with the server(s) **107**, for example. The software **108** is responsible for other functions, which are relevant to the operation of the unit, such as storing images and data to the device's **101** non volatile memory **104** in an encrypted or not encrypted, redacted or not redacted format, and controlling the content on the device's **101** display **119**, if available. The software **108** may also provide functions pertaining to the configuration, calibration, and operation of the device **101**. The software **108** may also communicate with the server(s) **107** for the purpose of downloading updates, settings, configuration parameter(s), neural networks(s), data and/or files. The software **108** may also communicate to the server(s) **107** of sending non-incident information, pertaining to the performance, status or diagnostic information for the device **101**.

(66) Referring to FIG. **3**, shown is a variety of options which the device **101** can be powered. A device **101** such as smartphone, smart camera or embedded computer system, with an optional battery **301** that is connected through a power connector **305** to a vehicle battery **307** or to a vehicle power supply **302** through a fuse panel **303** or auxiliary port(s) **304** directly or through a voltage converter **306**. The device **101** may have an optional ignition sense **308** port to know when to the device **101** on or off as to not drain its own battery **301** and/or the vehicle battery **307**. The device **101**, usually has a battery **301** that allows it to operate for a period or to keep it in a suspended state for a period of time without a direct connection to power. For example, nearly all smart phones today come equipped with a battery **301**.

(67) The device **101** typically connects to the vehicle's **102** power supply **302** through the vehicle's **102** fuse panel **303** or through auxiliary port(s) **304** which are typically known as automobile auxiliary power outlet, car outlet, automotive power socket, automobile outlet, or vehicular outlet. The auxiliary port(s) **304** may be with a USB connector, a plug connector, a socket connector, or other commonly used connectors. A vehicle's **102** power supply **302** voltage typically ranges from 9 v to 36 v. The device **101** may be a smart phone, a smart camera, or an embedded computer with varying power requirements. Depending on the compatibility between the device's **101** power requirements and the vehicle's **102** power supply **302** voltage, a voltage converter **306** may be required. The power load supplied by the vehicle's **102** electrical system would typically allow the device **101** to operate and charge its battery **301** simultaneously.

(68) Depending on the configuration of the vehicle **102**, when it is idle or turned off, power available to the auxiliary port(s) **304** may be shut off to protect the vehicle **102** battery from being drained. The device **101** can be connected to the vehicle **102** power through a power connector **305** with a compatible header. For example, if it is a smartphone the power connector **305** may be a USB-c, a lightning plug, or another charging cable variant. The other side of the power connector **305** may be a USB, a socket plug, a wire harness, or other connector that is aimed to connect the device **101** to the power supply **302** through the power system. The device **101** may also utilize proprietary or standard cables for power connector **305** for other device **101** variants. In the event the device **101** is an embedded computer, it may be connected directly or indirectly to the vehicle's battery **307**. It may also have an ignition sense **308** interface that will communicate to the device **101** when to turn off or on as to not to drain the vehicle's battery **307**.

(69) Privacy & Security

(70) Referring to FIG. **6**, shown is a privacy/security framework **600** with capabilities, i.e. as provided by the image processing instructions **905** for the redaction (e.g. generation of discard data **19**, **19′**) of personally identifiable information **601** such as faces **602**, license plates **603**, or other objects **12** containing personally identifiable information **604** such as cars or people. In certain circumstances the framework **600** has the option to redact whole image **605** other than object **12** of interest (also referred to as the resultant image data content **20**), depending upon a defined privacy criteria. The framework **600** has the option to utilize unredacted images **606**, with processing on device **101** and/or server **107**. Secure encrypted image **606** storage **104** on the device **101** and/or server **107**. Secure encrypted **606** communications to the server **107**. It is recognized that the images **16** once processed (examples as processed image results **601**, **602**, **603**, **604**, **605**, **606**) can be referred to as the processed image portions **20** (i.e. such that the data content of the processed image content **20** does not contain the discard image data **19**, **19′**).

(71) For example, discard data **19**, **19′** can include data such as but not limited to: portions of images **16** which are being blurred; images **16** which are determined to not contain incidents **12**; and/or images with incidents **12** which are being discarded due to their position (i.e. duplication in relation to an overlapping incident **12** as discussed above with respect to previous images **16** and newly acquired images **16**).

(72) As such, the image portions **20** contain less image data content than the unprocessed images themselves **16** (as acquired by the imager **500**). It is the image portions **20** (the result of processed images **16** as performed by the image processing instructions **905**) themselves that can be included as processed image data **20** in the object data **21** transmitted to the server **107** over the network **18**, for example. It is recognised that image **16** portions can refer to inter frames (i.e. some whole frames that are retained/dropped) or intra frames (i.e. area(s) within image **16** that are retained/dropped). The privacy/security framework **600** is one example of how the data of the images **16** can be reduced (by identifying and thus extracting the discard data **19**, **19′**) by the image processing instructions **905** before the object data **21** containing the objects **12** of interest are communicated to the server **107** or otherwise stored as processed image data **20′** in the storage **30** by the server **107**. It is recognized that an object **12** of interest included in the object data **21** can also be in redacted form (e.g. blurred out or pixel substituted), as such the included object **12** of interest in the resultant object data **20** can also include a reduction in its data size. It is recognised that pixel blurring can result in an increase in data size of the object data **21** (as compared to the same images **16** containing non blurred content). It is also recognised that pixel substitution/deletion can result in a decrease in data size of the object data **21** (as compared to the same images **16** containing non substituted/deleted content). Further, it is recognized that a particular image data **16** (e.g. image frame **16***a,b,c,d*—see FIG. **2**) can include redaction data **19**, **19′** (e.g. substituted or blurred image frame portion(s)) as well as object(s) **12** of interest. Accordingly, any particular image data **16**, prior to processing, can include only “to be determined” image discard data **19**, **19′** only “to be determined” object(s) **12** of interest, or both “to be determined” image discard data **19**, **19′** and “to be determined” object(s) **12** of interest. It is recognized that “to be determined” can also be referred to as “potential” or “candidate”, as desired.

(73) Considering that the system **10** is intended to be operated in the public space, and it is expected to be used extensively in public spaces and by governments and quasi-government organizations, the system **10** can have one or more privacy and security options intended address regulations regarding the collection and storage of data including personally identifiable information (pii) in the images **16**. In relation to data, pii may be pictures which uniquely identify individuals. Examples of personally identifiable information include images of people's faces **602** or of a vehicles' license plate **603**. It may also include house addresses. Sometimes, the objects **12** themselves may identify an individual, for example a fairly unique car or house. Governmental organizations typically have regulations and legislations related to the handling and storage of pii.

Images **16** acquired by governments are also typically subject to freedom of information requests. As such, in many occasions governments do not want to store pii. Governments and organizations responsible for maintaining assets are also typically subject to litigation and claims related to incidents causing property damage, personal injury, and/or death. Images **12** which are acquired by device **101** may occasionally be used as evidence against the system user in claims. As such, some users would only wish to retain the incident **12** data, but not any other information. For example, an image **16** may report an incident **12** of a pothole, but in the image **16** peripheral there may be a broken sign **12** not detected by the system. Such image **16** can then be used as an evidence against the owner of the information in claims. The system **10** can store an acquired image **16** containing as an incident **12** as an unredacted image **606** containing all of the images' **16** original information. However, the device's **101** software **108** can have options built in to exclude pii and other non-detected related data, using the privacy/security framework **600** s discussed above.
(74) Users opting to redact information may use the image processing instructions **905** to redact the whole image **605** (e.g. image **16**) other than the detected object(s) **12** of interest. The object(s) **12** of interest can be maintained in the picture **16**, whereas the remainder of the picture **16** can undergo image processing intended to redact the image **16** in order to generate the resultant processed image data **20**, **20′** and the discard data **19**, **19′**. The object(s) **12** of interest, can result in image data **20**, **20′** in which all image data **16** is redacted other than the object(s) **12** of interest. For example, a pothole **12** may be an object **12** of interest identified by the system **10**. After the image **16** redaction process, see FIG. **9** by example implemented as system **900**, the only object **12** that is not redacted in the resulting image data **20**, **20′** can be the pothole **12** itself.
(75) Users may also use the image processing instructions **905** to redact pii using object redaction **604** feature aimed to redact potential objects **12** containing pii, such as vehicles, cars and people. It may also only blur objects which are pii, such as license plates **603** and faces **602**. In this instance, only the pii objects **601** and/or objects containing pii **604** will be redacted (included in the discard data **19**) whereas the rest of the image **16** will remain untouched and thus such content being used as the resultant processed image data **20**, **20′**.
(76) The redaction operation of the image processing instructions **905** could obfuscate the details in the image **16** parts which are to be redacted (i.e. included in the discard data **19**, **19′**). Examples of redaction operations used to generate the discard data **19** can be pixel substitution and/or blurring. Pixel substitution is a process where pixels occupied by the object **12** of interest (whether such boundaries are semantic segmentation instances or bounding boxes) are replaced by pixels of a single color or a pattern of colors, or otherwise treated as an absence of image data **20**, **20′**. A blur is a visual effect function that makes the details in resultant image data **20** appear fuzzy or out of focus. The redaction operation can take place on the device **101** or on the server **107**, depending on whether the first image processing portion **402** or the second processing image portion **404** is utilized (see FIG. **4**) for the particular redaction operation.
(77) For example, referring to FIG. **4**, the first image processing portion **402** can be used to generate discard data **19** (using pixel substitution) and the resultant image data **20** (that includes object(s) **12** of interest), such that the resultant image data **20** is included in the object data **21** and transmitted to the server **107**. Once received, the server **107** can use a set of image processing instructions **905** to implement pixel blurring and/or additional pixel substitution, thus generating further discard data **19′** as extracted from the resultant image data **20** of the object data **21**. The server **107** would then store modified object data **21′** in storage **30**, which includes the resultant image data **20** other than the extracted discard data **19′** (see FIG. **1**).
(78) In the event that the resultant images **601**, **604**, **605**, **606** and their associated data are stored on the internal volatile memory **104**, the resultant images **601**, **604**, **605**, **606** and data may be encrypted using a modern encryption algorithm by the image processing instructions **905**, which would obfuscate the files **706**. Therefore, in the event that the device **101** is stolen from a vehicle **102** or lost, the information (e.g. object data **21**) stored on the device **101** would not be easily

accessible. Communication of the object data **21** to the server **107** may place over encrypted communications to ensure that the data is secure in transit. Finally, stored information (e.g. object data **21**) may be encrypted on the server **107** to ensure that the data is secure while at rest.

(79) Referring to FIG. **7**, the device **101** can have a variety of sensors **700**. In addition to camera(s) **500**, the device **101** is equipped with a geo-positioning **701** sensor. The device **101** may also include an accelerometer **702** sensor, gyroscope **703** sensor, rotational vector **704** sensor, and other sensors **705** that can provide information **17** regarding the movement, position and/or orientation of the device **101** and/or the vehicle **102** on which it is equipped. The sensor(s) **700** and camera(s) **500** data **17**, **16** is processed by the software **108** (including the image processing instructions **905**) before being sent to the server **107**, as discussed herein. The sensor data **17** and image data **16**, **20** may be stored on the device **101** non volatile memory **104** in the form of file(s) **706** or database **707** entries prior to being transmitted to the server **107** as object data **21**.

(80) The device **101** includes a geo-positioning sensor **701** to determine its geo-spatial coordinates **17**. Geo-location, or a geo-positioning sensor **701**, provide location based information **17** using satellite (such as GPS, glonass, galileo) or cellular tower locations to determine device positioning information, which is associated with the images **16** (e.g. geo coordinates GCa, b, c, d—see FIG. **2**). In addition, the device **101** in many instances will have additional sensors **700**. For example, modern smartphones **101** on the market today have a variety of sensors **700** embedded right onto them, which provide information that can be used to determine the device **101** orientation, pitch, magnetic pole direction, geo-spatial position, velocity, acceleration, shock, vibration and other data **17** related to position and movement. The device **101** may include an accelerometer sensor **702** used to measure the acceleration force **17** applied to the device **101** across its x axis, y axis and z axis. The force may or may not include the force of gravity. The acceleration data **17** will typically be available as meters per second squared (m/s.sup.2) though it may be in other units (for example voltage) that can be converted to such units. The device **101** may include a gyroscope sensor **703** used to measure the rate of rotation across the device's **101** x axis, y axis and z axis. The gyroscope data **17** will typically be available as radians per second (rad/s) though it may be in other units (for example voltage or frequency) that can be converted to such units. The device **101** may include a rotational vector **704** sensor used to measure the degree of rotation across the device's **101** x axis, y axis, z axis and an optional scalar product or quaternion. The rotational vector data **17** will typically be degrees, though it may be in other units (for example voltage) that can be converted to such units. The device **101** may include other sensor(s) **705** to measure a variety of other conditions **17** related to the movement, acceleration, forces applied and position of the device **101**. For example, the device **101** may include a gravity sensor **700**, which would measure the force of gravity **17** in relation to the device **101**. Other sensor(s) **700** may also be magnetometer which can determine the device's **101** position **17** in relation to the magnetic north or true north. Other sensor(s) **705** may also include hardware and/or software monitoring of the device **101** components. Examples of sensors **700** can include battery level sensor, battery temperature sensor, cpu temperature sensor, gpu temperature sensor, ambient temperature sensor, cpu core utilization, cpu overall utilization, luminance sensor, proximity sensor, and other built in sensors available for the device **101**.

(81) Any and all of the above discussed sensor type data (i.e. sensor data **17**) can then be associated with camera(s) **500** images in order to determine additional insights. For example, the sensor data **17** may be used to derive ridership experience, level of vibration, the speed in which the vehicle **102** is travelling, whether the device **101** is within a geo-zone, or the estimated geo-positioning of an object **12** detected in an image **16** in relation to the device **101**. The sensor data **17** may also be used to optimize the performance of the device **101** in relation to the current heat, power and processing situation.

(82) The sensor(s) **700** and camera(s) **500** provide for data **17**, **16** to be acquired and processed by the software **108**. The resultant processed data **16**, **17** (e.g. using the first data processing portion **402**) is then either transmitted to the server **107** or stored on the device **101** non volatile memory

104 until transmission can take place. The data **16**, **17**, **20** may be stored as file(s) **706** in variety of formats, such as xml, csv, txt, or in a proprietary format. The data **16**, **17**, **20** may also be stored in a database **707**. The data **17**, **20** may be stored and transmitted in encrypted on non-encrypted format.
(83) The data **17**, **20** may be further processed on the server **107** using the second data processing portion **404**. For example, it may be correlated with road segments, assets, and other information to derive additional insights. For example, what roads or assets were inspected. It may also be used by the server **107** for detecting alerts related to device constraints (heat, power, or processing capabilities) on the device **101**.
(84) The GPS receiver (i.e. position sensor **701**) can be used to record (sensor data **17**) the location coordinates GC a, b, c, d where the incident **12** occurred so that the location of the incident **12** can be presented. GPS can also be used by the instructions **905** to determine the speed the vehicle **102** is travelling which is used to activate speed enabled features within the application (e.g. software **108** features of Screen lock, driver attention warning). In addition, the GPS data **17** can be used as breadcrumbs' to track the road surfaces **14** that have already been inspected, thus when the instructions **905** are used to compare the geo coordinates GC of the newly acquired images **16**, those images **16** being determined as duplicates (i.e. having matching geo coordinates GC to previously acquired images **16**) can be discarded/excluded from the object data **21**. Furthermore, GPS data **17** can used by the software **108** to determine if roads **14**/surroundings **13** being scanned are within a defined geofence zone, otherwise detections **12** will be ignored.
(85) The accelerometer and magnetometer sensors **700** can be used by the software **108** to determine the vehicle's **102** direction of travel. Being able to determine when vehicle **102** direction of travel is necessary for the system **10** to present which side of the road (e.g. North, east, west or south bound lane) is being scanned by the system **10**. It can also help to identify the incident **12** location in relation to the device **101** or the vehicle **102**.
(86) In view of the above, the rotational vector sensor **704** can be used to determine the device **101** orientation, including pitch, facing direction, vibration and bumps and such information **17** is sent to the server **107** together with the image data **20**. Further, it is envisioned that the device **101** can keep track of rotational vector sensor **704** information **17** for the purpose of integrating the data with data obtained from images **16** for the purpose of detecting road quality and road roughness levels as determined by the level of “vibration” or “bumps” detected by the sensor **704**, and such data is sent to the server **107** for the purpose of being correlated to the image data **20** uploaded.
(87) The accelerometer **702** sensor can determine the device's **101** acceleration force along its x axis, y axis, z axis and such information **17** is sent to the server **107** together with the image data **20**. The gyroscrope sensor **703** sensor can determine the device **101** orientation **17**, and such information **17** is sent to the server **107** together with the image data **20**. The other sensors **705** can be such as magnetometer **705** sensor to determine the device **101** orientation **17**, and such information is **17** sent to the server **107** together with the image data **20**. The device **101** can keep track of the location, via the sensor(s) **700**, in which the device **101** was present through gps breadcrumbs or routes as evidence that the device **101** inspected the area.
(88) Referring to FIGS. **4** and **8**, one example of a distributed processing system **400** is a first image processing portion **402** for image pre-processing, which is the process of manipulating and/or adjusting the image **16** data acquired by the camera **500** to provide that the data received for neural network **905** processing is inputted in a way which will optimize the results and performance of the system **10** (i.e. optimize the quality and quantity of the resultant image data **20** in the object data **21**).
(89) For example, using the software **108**, a resolution for the image **16** data can be selected from one of the camera's supported resolutions **801**. The camera resolutions **801** can be represented as a name, such as 8k, 4k, 1080p, 720p, hd, or other common names. It can also be represented as a resolution, representing the number of pixels and typically in a format of width×height, for example 7680×4320, 3840×2160, 1920×1080, 1270×720, or other resolutions. In many instances,

neural networks **905** can be optimized to accept images **16** in certain resolution, typically referred to as "input shape". For example, an image **16** can be acquired by the camera **500** at a resolution of 1080p (1920 pixels×1080 pixels). However, the neural network **905** model can be trained on images **16** scaled down to the size of 300 pixels×300 pixels. The software **108** then needs to resize, or adjust the resolution **801** of the image **16** dimensions from 1920 pixels×1080 pixels to 300 pixels×300 pixels in order for the neural network **905** to process it appropriately. The neural network **905** model can have a different input shape than 300 pixels×300 pixels it may be higher (for example, 600 pixels×600 pixels) or smaller (for example, 224 pixels×224 pixels). Typically, the larger the model "input shape" or resolution, the slower the images **16** will be processed, however the larger the "input shape" resolution is, the more details will be retained in the image **16** which may increase the model's effective detection parameters, such as accuracy, recall, precision, f-score and other such metrics.

(90) The software **108** can also facilitate for a field of view adjustment **802**, the field of view adjustment **802** can be optical zoom level adjustment, if supported by the peripheral camera **500** of an embedded computer system **101**. If further magnification is required to calibrate the camera's **500** field of view, the digital zoom level can be adjusted through the software **108** to achieve the desired optimal field of view. The software **108** may also select from a variety of internal camera(s) **501** or external camera(s) **502** in order to adjust the field of view **802**. Different field of view may be optimal for different use cases on vehicle(s) **102**. For example, the height and pitch in which the camera **500** is mounted may be different for a bus, a service truck, or a sedan, and may require different zoom levels in order to cover the same number of lanes. Similarly, different field of views may be preferred for different objects **12**. For example, signs **12** may favor a wider field of view that covers the surroundings **13** whereas road defects **12** may favor a narrower field of view covering the road surface **14**.

(91) The software **108** can also facilitate for cropping **803** parts of the image **16**. Cropping the image **16** allows to omit areas which typically do not require neural network **905** processing. For example, if the camera **500** is mounted on a windshield, the top 20% of the image **16** may typically be sky, whereas the bottom 10% of the image **16** may be a dashboard or a hood of a vehicle **102**. By cropping parts of the image **16** which are not relevant, those areas are less likely to generate false detections **12** or false incidents **12**. In addition, in the event that the neural network **905** "input shape" is lower than the acquisition resolution, by cropping out irrelevant portions of the image **16**, less detail is lost in the resizing operation of the image **16**. Cropping **803** may also be used for extracting a portion of an image **16** for additional image processing and/or inference activities **905**. For example, a car **102** may be detected **12** by a neural network **905** and then cropped from the image **16**. The cropped car **102** may then be either redacted **604** or processed through a neural network **905** that is trained to identify license plates in a picture **16**. Operations related to redaction such as personally identifiable information redaction **602**, **603** object redaction **604** and/or image redaction **605** are also considered image processing operations as performed by the software **108** and related instructions **905**.

(92) In the event that the image **16** data acquired by the device's **101** camera(s) **500** is in a format that is not compatible or optimized to be used with the neural network **905** architecture or library, color space conversion **804** may be required. Examples of color spaces include yuv, rgb, hsv, hsl, cmyk and others. Even within color spaces, there are variations in the container, structure, channels, order, format, decimal system which may require conversion. For example, a file may be represented in bytes, words, or hexadecimal. Another example is that rgb channels may be ordered as bgr. Another example is that extra channel may be present to represent transparency (rgba).

(93) Once the image data **16** is preprocessed by the first image processing portion **402** by the device **101**, e.g. using image processing instructions **905**, then the resultant object data **21** would be sent to the server **107** for implementation by the server **107** of image processing instructions **905** during a second image processing portion **404** by the server **107**, which then results in the resultant

object data **20′** being stored in the storage **30** (see FIG. **1**). As such, it is recognized that the device **101** can have the image processing instructions **905** as an embodiment of the system **10**. As such, it is recognized that the server **107** can have the image processing instructions **905** as an embodiment of the system **10**. As such, it is recognized that both the server **107** and the device **101** can have the image processing instructions **905** as an embodiment of the system **10**. The device **101** and the server **107** can have different variations, configurations, and parameters relating to the image processing instructions and neural network(s) **905** used.
(94) Referring to FIGS. **1** and **9**, shown is an example of image processing system **900** implemented by the software **108** (including the image processing instructions **905**) on the acquired images **16**, in order to produce the processed image data **20** to be included with the object data **21** transmitted to the server **107**. It is also recognized that the example of image processing system **900** implemented by the software **108** (including the image processing instructions **905**) on the resultant image data **20**, in order to produce the processed image data **20′** to be included with the received sensor information **17** stored in the storage by the server **107**.
(95) The software **108** can include image instructions **905** (e.g. including artificial intelligence neural networks **905**), for image **16** processing and inference for flexible workflows **906** inclusive of neural network(s) **905** inference operations **907** including detection **902**, classification **903**, and segmentation **904**, in order to generate the discard data **19**, **19′** as well as the resultant processed image data **20**, **20′**. It is recognized that the workflows **906** can include a plurality of different numbers/combinations of the operations **907** in any order, as configured in the image processing instructions **905**, in order to identify, classify and segment any object(s) **12** in the image(s) **16** under consideration. The system **900** also depicts processing **800** of images **16** containing object(s) **12** of interest in relation to incidents. One image **16** may have several different workflows **906** applied to it. The object(s) **12** of interest are also referred to as classes **12**. The class **12** refers to one of the output categories for the object(s) **12** of interest. For example, they may include but are not limited to: pothole **12**, car **12**, person **12**, sign **12**, etc. The network(s) **905** can detect, classify, and/or segment one or more classes **12** (also referred to as object(s) **12** of interest) in the image **16**.
(96) It is recognized that the identified object(s) **12** of interest are included in the processed image data **20** while the discard data **19** is excluded from the processed image data **20**, as one embodiment, such that the processed image data **20** and the sensor data **17** is transmitted to the server **107** as object data **21**.
(97) Further, it is recognized that the identified object(s) **12** of interest are included in the processed image data **20′** while the discard data **19′** is excluded from the processed image data **20′**, as one embodiment as implemented by the server **107** using the object data **21** obtained from the device **101**.
(98) Further, it is recognized that the identified object(s) **12** of interest and discard data **19** are included in unprocessed images **16** sent to the server **107** by the device **101** as the object data **21** (including the sensor data **17**). Once received, then the server **107** would then process the images **16** as processed image data **20** while the discard data **19** is excluded from the processed image data **20**, as one embodiment as implemented by the server **107** using the object data **21** obtained from the device **101**.
(99) Typically, image(s) **16** acquired by the device's **101** camera(s) **500** are available in some initial resolution, color space, and formatting. It is expected that in many cases, the image(s) **16** may need to undergo image processing **800** operations to optimize their compatibility with the neural networks **905** used and the object(s) **12** of interest which they are trained to identify. Some examples of image processing **800** operations are resizing or adjusting resolution **801**, field of view adjustments **802**, cropping **803**, and/or color space conversion **804**, as shown in FIG. **8** by example only.
(100) As such, the image processing **800** operations can include the resolution of the image **16** can be set based on the available resolutions present on the camera **500** device, whether available as

resolutions or as a name representative of the resolution. Further, the field of view can be adjusted via adjusting the optical zoom levels of the camera(s) **500**. Further, the field of view can be adjusted by a digital zoom process, wherein the picture **16** is magnified and only the parts of the image **16** that remain within the original dimensions are processed. Further, the region of interest **12** in the image **16** can be set. Once set, the region of interest **12** will be cropped. Further, the image processing can include color space conversion, whether from one space to another, or adjusting the formatting, order and/or channels of the utilized color space.
(101) For example, the processing instructions **905** (e.g. neural network **905**) can be defined as a set of functions, operations and/or instructions which facilitates for the system **900** to train itself based on annotated datasets, commonly referred to as “ground truth”. Once trained, the system **900** can then infer on new datasets. The process is known as machine learning. The neural network(s) **905** utilized in the system **900** can be primarily geared towards identifying object(s) **12** of interest in images **16** for automated incident identification and reporting. Once processed using the image processing instructions **905**, the system **900** outputs the processed object data **20**, shown by example in FIG. **6**. Further, software **108** is configured to construct the object data **21** by associating the sensor information **17** (e.g. including geo coordinate data GC a, b, c, d) for each of the objects **12** of interest. It is recognised that during the processing of the images **16** using the image processing instructions **905**, some of the image **16** data acquired will be discarded in view of the discarded image **16** data does not contain any objects **12** of interest as determined by the image processing instructions **905** as executed by the processor **111** and/or GPU **112** of the device **101** (see FIG. **7**). It is recognized that discarded image **16** data can be referred to as discarded data **19** (see FIG. **4**), such that discarded data **19** is not included in the object data **21** and is thus can be inhibited from being transmitted over the network **18** to the server **107**.
(102) The neural network(s) **905** utilized can have a plurality of architectures which pass the image **16** through a sequence of layers operations **907** which are aimed at aggregating, generalizing, manipulating and/or modifying the information of another layer for the purpose of inferring, detecting, classifying and/or segmenting objects **12** in images **16**. Examples of some typical operations in neural network(s) **905** are: (a) convolution; (b) rectification; (c) fully connected; (d) pooling layer (e) bottleneck and/or (f) loss layer.
(103) The architecture of the system **900** can be a neural network **905** architecture such as: (a) single shot detector (ssd), (b) you only look once (yolo), (c) convolutional neural network (cnn), (d) region-based convolutional neural network (rcnn), (e) fast region-based convolutional neural network (fast rcnn), (d) faster region-based convolutional neural network (faster rcnn), (e), mask region-based convolutional neural network (mask-rcnn), (f) region-based fully convolutional networks (r-fcn), or other published neural network **905** architectures.
(104) When a neural network **905** is trained on an image **16** set (e.g. a series of image frames **16***a,b,c,d*), it can set certain parameters commonly known as weights. The parameters, or weights, are typically stored in a model file, or weights file. The neural network **905** utilized in the system **900** can be trained using published, well known, weights files as the basis. For example, mobilenet (such as mobilenetv1, mobilenetv2, mobilenet v3), inception (such as inception v1, inception v2, inception v3), vgg, or other popular pre-trained networks, and can be composed of different number of layers (for example, resnet50, resnet101). However, the concept of such pre-trained neural networks **905** is the same whereas a base architecture with base weights is modified whereby one or more of the last or final layers is modified to detect or classify a set of objects **12** of interest, which may be identical, exclusive, partially inclusive, or fully inclusive of the original trained objects and may include new objects not present in the original neural network **905**. Neural network(s) **905** may also be of a proprietary custom architecture with weights or parameters which are trained from scratch.
(105) The neural network(s) **905** may be utilized as a detector **902**, see FIG. **9**. A detector **902** typically identifies an object **12** of interest in image(s) **16**, and the location of the object **12**. The

location of the object **12** is typically in the form of a bounding box represented by coordinate(s) REFa, b, c, d and/or distance(s) in relation to a point of reference **902** in the image **16**, see FIG. **2**. A detector **902** may also provide a score, typically known as confidence, which represents how sure the neural network **905** is in the object **12** detection. A detector **902** may also detect landmarks. Landmarks are points of reference in a known object **12**. For example, in the context of a detector identifying a sign, the bottom of the sign pole and the top of the sign pole may be landmarks. Such landmarks can then be analyzed to derive further information about the status of a sign—for example, whether it is crooked or not.

(106) The neural network(s) **905** can be utilized as a classifier **903**. A classifier **903** has a list of potential classes, or object types, which it is trained to identify in a picture. When processing image(s) **16**, a classifier **903** typically returns a list of potential object(s) **12** in the image **16**, sorted by the model's confidence of their presence in the image **16**. The neural network(s) **905** can be utilized as a segmentor **904**. A segmentor **904** typically segments image(s) **16** into regions. The regions are then typically predicted to belong to a certain class **12**, or type, which allows to extract a mask, or a pixel blob, that represents the class **12**. A segmentor **904** can also separate instances of the object(s) **12** into separate object(s) **12** representing one or more classes **12**. For example, a segmentor **904** may identify a pothole **12**, and also the shape of the pothole **12**, which will allow to estimate its surface area and severity.

(107) The neural network(s) **905** can be designed and/or optimized to be used on the device's **101** gpu, cpu or both. The workflows **906** may utilize one or more neural network(s) **905**, and the neural network(s) **905** may be used in a sequence. One neural network(s) **905** can responsible for detecting **902** objects and/or regions of interest in the image(s) **16**, and one or more additional neural network(s) **905** can be responsible for classifying **903** the objects **12** and/or regions of interest already detected in the image(s) **16**. For example, a neural network **905** may detect **902** a pavement crack **12**, crop it with image processing **800**, and then another neural network **905** classifies **903** it as a longitudinal type of crack **12**. It could also be used to verify that the first detection is correct. For example, the first neural network **905** may detect **902** a pothole **12**, crop it using image processing **800**, and pass it to a classifier **903** which confirms it is a pothole **12** and not a manhole. In some situations, this process provides the opportunity to classify **903** the object **12** of interest using a higher resolution, since the detector **902** may infer on a scaled down version of the image **16**, whereas the cropped image **16** would be inferred at a higher resolution.

(108) One neural network **905** can be responsible for detecting **902** objects **12** and/or regions **12** of interest in the image(s) **16**, and one or more additional neural network(s) **905** is responsible for detecting **902** additional objects **12** and/or regions **12** of interest in the already detected area(s) of the image **16**. For example, a neural network **905** detects a car **12** and then another neural network **905** detects a license plate **12** on the cars **12**. One neural network **905** can be responsible for detecting **902** objects **12** and/or regions **12** of interest in the image(s) **16**, and one or more additional neural network(s) **905** can be responsible for extracting landmarks **902** from the objects **12** and/or regions **12** of interest in the image **16**. For example, a neural network **905** detects a pothole **12**, and then another neural network **905** will identify its topmost point, bottom-most point, leftmost point, and rightmost point, and return those in a coordinate format respective to the image **16**, or in a coordinate format respective to the object/region **12** of interest.

(109) Further, the neural network inference can be processed on the Device GPU (in addition to the CPU **111** resident in the computing infrastructure **100**. The neural network **905** can infer multiple classes **12** simultaneously. Further, one or more of the neural networks **905** can be simplified by approximating the neural network to floating-point numbers for the purpose of reducing the memory and processing requirements. Such reduced neural networks, sometimes known as Quantized neural networks, are then used on the Device **101** CPU **111**.

(110) As discussed, the image processing instructions **905** can include utilizing sensor data **17** to interpret/decide upon objects of interest **12** and/or discard data **19**. For example, sensor data **17**

(such as GPS data associated with the images **16**) can be used (as configured in the image processing instructions **905**) by the device **101** (and/or server **107**) to discard a portion/whole image frame **16***a,b,c,d* from inclusion in the object data **21**.

(111) Referring to FIG. **10**, shown are example inference results **1000** obtained from the image processing instructions **905** (e.g. neural network(s) **905**). Neural Network(s) Inference results **1000** of a processed image **16**, including bounding boxes **1002**, polygons **1003**, masks **1004**, and landmarks **1005** as part of the resultant processed image data **20**. When the system **900** is processing an image **16** through a Neural Network **905**, it returns inference results in the form of data **20**. The data **20** is typically a list or array of things **12** it is trained to find, typically known as classes12 and typically represented as a class id which can be correlated with a name (i.e. class 1 is person, class 2 is car, etc) and numerical score. The score typically represents the Neural Network's **905** confidence in the class **12** as identified in the image **16**.

(112) The Neural Network **905** can also provide additional information, per object **12**, as to where the object **12** is found in the image **16** in the form of a Bounding Box **1002**, which is typically a rectangle that is encompassing the object **12**. The Bounding Box **1002** information could be provided in a variety of formats which could be used to construct a rectangle. For example, it could be two opposing coordinates (i.e. top left, bottom right) in the rectangle, or a center coordinate provided also with width and height parameters. The Neural Network **905** can also provide additional information, per object **12**, as to where the object **12** is found in the image **16** in the form of a Polygon **1003**, which is typically a series of connected points that are encompassing the object **12**. The Polygon **1003** information could be provided in a variety of formats which could be used to construct it.

(113) The Neural Network **905** can also provide additional information, per object **12**, as to key features of the object **12** in the form of Landmarks **1005**, which is typically one or more points which are expected to be present in the object **12**. Landmarks **1005** could represent many things. For example, landmarks could represent lane markings **12** and edge **12** on a road. They could represent the top and the bottom of a sign post **12**. They could represent the top and bottom edge of a pothole **12**. Neural Network(s) **905** can learn to identify any landmark(s) **12** in any object **12**. The coordinates for the Bounding Boxes **1002**, Polygons **1003**, and Landmarks **1005** may be absolute pixel coordinates in the image **16** (in relation to top left corner of the image), they may also be relative to another point in the image **16**. The coordinates may also be provided in the form of a fraction, or percent of the image **16**, which could then converted to the pixel representation.

(114) The Neural Network **905** can also provide additional information, per object **16**, as to where it is found in the image **16** in the form of a Mask **1004**, which is also known as instance segmentation. The Mask **1004** information typically maps, on a pixel level, which pixels in the image belong to a specific object **12**. The system **900** can identify one or more object **12** types, or classes **12**, in the same image **16**, using one or more Neural Network(s) **905**. For example, it may identify signs **12**, potholes **12**, people **12** and cars **12**.

(115) Since the system **900** facilitates automated incident **12** detection in the images **16**, the Neural Network(s) **905** can have the option automatically identify incidents/objects **12**, including such as but not limited to: (a) cracks, including some or all of the following crack types: longitudinal crack(s), linear crack(s), transverse crack(s), reflection crack(s), lane joint crack(s), widening crack(s), fatigue crack(s), alligator crack(s), crocodile crack(s), block crack(s), hair crack(s), edge crack(s), edge joint crack(s), slippage crack(s), and/or shrinkage crack(s). The cracks may be further differentiated by severity, such as the extent of the road to which the cracks apply and how wide the cracks are; and/or (b) deformations and/or distortions in the pavement, including some or all of the following: rutting, shoving, corrugation, depression, and/or upheaval; and/or (c) road repairs, including some or all of the following: asphalt patching, pothole patching and/or crack sealing; and/or (d) road damage, including some or all of the following: potholes, stripping or ravelling. (e) street signage issues including some or all of the following: unreflective signs, bent

signs, faded signs, obstructed signs, damaged signs, crooked sign, twisted sign, and/or vandalized sign; and/or (f) manhole issues, including some or all of the following: such as raised manhole, sunken manhole, and/or open manhole; and/or (g) drainage related issues, including some or all of the following: water pooling on pavement and clogged catch basins; and/or (h) pavement marking issues, including some or all of the following: faded markings, and unreflective lane markings; and/or (i) road obstruction issues, some or all of the following: debris on road, cadavers, and construction garbage bins/dumpsters which are obstructing the right of way; and/or (j) side walk issues, including some or all of the following: trip edges, cracks, chips, distortion and/or deformation.

(116) In addition to identifying and reporting incidents **12** automatically, the Neural Network(s) **905** can also infer for other purposes as well, such as but not limited to: (a) The Neural Network(s) **905** may assess environmental conditions which may affect the system functionality, including some or all of the following conditions: heavy rain, heavy fog, heavy snowfall, occluded windshield, daytime lighting conditions, night time lighting conditions, reflections on the windshield, and/or sun glare. The Device's **101** software **108** may then adjust its algorithm and parameters to optimize the system's performance under the identified environmental conditions. For example, an incident identified while the Vehicle's **102** windshield washer is in operation, the image would be discarded if the fluid spray blocked the field of view. Another example, if the road **14** is classified as completely covered in snow, pothole **12** detections would be excluded. The primary purpose of the environmental neural network **905** is to reduce erroneous incident occurrences, but it could also be used to report conditions and incidents. For example, if the road is covered in snow, it may be due the fact that the snow plows have not covered that road, which may be considered as an incident. (b) The Neural Network(s) **905** may assess the road **14** type in order to know which type of models, workflows, algorithms or Neural Network(s) **905** to apply. For example, the road **14** may be asphalt, concrete, gravel, dirt, trail or other types of roads. (c) The Neural Network(s) **905** may assess the road **14** conditions for the purpose of collecting road ratings for the roads which are travelled (such ratings may also be known as Pavement Condition Index). Examples of ratings may be a quantitative numerical value between 0 to 100, or a descriptive qualitative rating such as excellent rating, very good rating, good rating, fair rating, poor rating, fail rating, failed rating, failure rating, and/or other ratings descriptive of the pavement condition. (d) The Neural Network(s) **905** may identify object(s) **12** of interest representing assets which are in a good state, whereby the absence of those objects **12** over time may be an incident. For example, street signage (which can be damaged or dislocated by high winds), traffic signals (which can be out of order), street lights (which may be burnt), lane markings (which can fade). The presence of such objects **12** could be correlated with a GIS database and ruleset to generate absence incidents. If an asset has not been detected for a certain amount of time, it gets flagged in the system. For example, this could be used to indicate that a stop sign has potentially blown down due to high winds or that a speed limit sign has been obstructed by vegetation. (e) The Neural Network(s) **905** may identify object(s) **12** of interest representing object(s) which are to be redacted for privacy.

(117) Referring to FIG. **11**, show are example display options for a display **119** including view options such as an incident image **1101**, user interface **1102**, viewfinder **1107**, safety mode **1103**, navigation mode **1104**, and a background mode allowing third party apps to run **1105**. The device **101** can have a display **119** which may be integrated or external. For example, a smart phone or a smart camera **101** can have an integrated display whereas an embedded computer **101** may or may not have it, depending on its configuration. The device's **101** software **108**, depending on the programming, may display different options.

(118) The device **101** may have a user interface **1102** screen that will allow the vehicle's **102** operator to choose different options pertaining to the operation of the system **10**. The user interface and its settings, options and/or menus can be accessible through a touchscreen, built in button and/or a remote control. Typically, when the device **101** is initially installed or operated in a vehicle

102, a viewfinder **1101** option will be enabled, which will allow the vehicle **102** operator to know that the device's **101** cameras **500** are aligned properly. The display **119** of the device **101** can be configured to minimize the distraction to the driver by turning off the screen or by displaying a warning message to not operate the device **101**, when the vehicle **102** is in motion or even by disabling the user interface to prevent driver from interacting with the device when the vehicle **102** is travelling faster that a configurable threshold.

(119) The display **119** can also be configured to assist the driver during operation by displaying a navigation **1104** screen. The navigation screen **1104** may show the device's **101** current position, and/or highlighted routes of roads **14** to patrol via a map interface **1104**. The device's **101** software **108** may also run as a service in the background of the device **101**, allowing a third party application **1105** to run on the device **101** while the software **108** is running in the background. For example, once the software **108** is started, other applications related to navigation, automated vehicle **102** location, work order management, dashcam video recorders, and other applications could be launched in the foreground. The third party applications may be launched by the software **108** and/or by the device's **101** operator.

(120) Referring to FIG. **12**, shown is an example incident/object **12** identification and reporting operation **1200**. Depicted by example is Vehicle **102** equipped with a Device **101**. The Device **101** identifies object(s) **12** of interest which are to be reported **1201**. The Device **101** then transmits data **1202** to the Server(s) **107**. The Server(s) **107** have files **1203**, one or more database(s) **1204**, and Software(s) **1205**. The Server(s) **107** communicate with Client(s) **1208** (which may be the devices **101** or separate computing devices with a browser or software that provides access to the system **1200**) which allow users to handle the incidents **12** using a user interface **1209** which may present the incidents **12** in a variety of ways including a map view **1210**, list view **1211**, and gallery view **1212**. Object(s) **12** of interest identified and reported **1201** pertains to possible inference results (e.g. resultant processed object data **20**, **20′** and associated sensor data **17**) as described in further detail in FIG. **10**—Neural Network(s) Inference Results **1000**.

(121) Further, for example, data transmitted **1202** illustrates what typical incident data **12** contains, which can include some or all of the following: (a) IDs, which may include the Device **101** ID, image ID, detection ID, user ID, and/or other ID's allowing to associate the data; (b) Longitude: east—west position of a point on the Earth's surface. Different countries may use different names or system to describe the point; (c) Latitude: north—south position of a point on the Earth's surface. Different countries may use different names or system to describe the point; (d) Sensor(s) Data **17**: described in greater detail with respect to FIG. **7**—Sensors **700**; (e) Detected Object(s) Data: described in greater detail in Neural Network(s) Inference Results **1000**; (f) Date & Time: a date and time signature pertaining to the time and date that the incident was identified; (g) Image: a picture of the incident capture by the Device's **101** camera(s) **500**. The image may be redacted; and/or (h) Other data: Other data pertains to other data that may be derived by the Device's **101** software **108** from its configuration, settings, sensor(s), database(s). For example, the direction to which the Vehicle **102** is travelling may be determined based on the Device's **101** sensors **700**. Other examples of other data may include the road segment on which the incident was obtained, or the direction that the Vehicle **102** was facing, or a geo-zone in which the incident was obtained.

(122) In view of the above, the incident data **21** is transmitted to the Server(s) **107** where it is processed by the Server(s) software(s) **1205** and organized and stored in database(s) **1204**. Some data, such as uploaded incident images **20**, **16** containing the incidents **12**, may be stored in the form of Files **1203**. The server **107** also provides for client(s) **1208** to securely log in to access a user interface **1209**, which may be either a web application that can be accessed using a web browser or a client/server application that uses physical installation to a computer or a smartphone **101**. Through the user interface **1209**, clients can view incidents **12** detected by the device **101**, which have already been uploaded to the server **107**, which can be visualized in a variety of ways, such as but not limited to: (a) a Map View **1210** which depicts a map, along with pins representing

detected incidents and known assets; (b) a List View **1211** which depicts a table along with fields related to each incident; and/or (c) a Gallery View **1212** which depicts a gallery of incidents in the form of clickable images.

(123) Further, by example, users can click on the pins to display more information **1214** about the incident **12** such as image, details of the detection, severity, the date and time the detection occurred. Through a click of a button, the detected incidents **12** can also be presented in a gallery view layout, where a thumbnail image of each incidents can be presented in a grid.

(124) Client(s) **1208** pertains to software that is used to access the system **10**. It can typically be a web browser, but it may also be a dedicated desktop application and/or a smartphone app. The Client(s) **1208** user interface **1209** may have different views to present the data the user. For example, when selecting the incident **12** or asset for which it is related in the appropriate view, more information is displayed to the user.

(125) The Server(s) **107** may process the information (object data **21**) in a variety of ways, for example, the Server(s) **107** can also associate incidents **12**, through their GPS coordinates, to a road **14** network segment, which is a representation of a segment of a road **14**, which typically includes geospatial and descriptive data, such as points, features and other fields—for example the class of the road (highway, local, regional), the street name, and/or the address range which it covers. Road segments are typically extracted from a geospatial database such as a shape file, KML file, KMZ file, XML, GML and/or other such popular formats used for the modelling, transport and storage of geographic information.

(126) The Server(s) **107** may also have an asset database, particularly for road signs, manholes, and catch basins, where the GPS coordinates, direction of travel and type of asset are logged in for every detected asset when they are detected on the Device **101**. This database can be used for inventory purposes or as a list for manual inspections.

(127) The GPS coordinates **17** and sensor information **17** of the incident **12** or asset may further processed on the server to determine additional insights, including some or all of the following: 1) Nearest address; 2) Nearest road intersection; and/or 3) Nearest major road intersection.

(128) It is recognised that the server **107** can be a physical server connected to the internet **18**. Alternatively, the server **107** is a virtual server connected to the internet **18**, and whereas it may be hosted on one physical machine or on a server cluster. Alternatively, the server **107** is cloud based and connected to the internet **18**.

Claims

1. A system for collecting, processing, discarding, storing, transmitting, and disseminating data, the data includes information captured from one or more cameras and one or more sensors mounted in a vehicle, and generated object data and generated resultant data, the system comprising: at least one device per vehicle, the at least one device having: a housing; one or more processors for executing software instructions; one or more memory components; one or more network interfaces; the one or more sensors for acquiring sensor data; the one or more cameras for obtaining digital images; a mounting component for mounting the housing of the at least one device to the vehicle; and the software instructions, including image processing operations and related software instructions to discard image data, generate the object data and the resultant data; and one or more servers, communicating with the at least one device over a communications network, the at least one device mounted in one or more vehicles.

2. The system of claim 1, wherein the object data and resultant data are transmitted to a server of the one or more servers using a cellular or wireless network as the communications network.

3. The system of claim 2, wherein processing of the digital images, the object data and the resultant data is performed by the at least one device alone, by the server alone, and/or by both the server and the at least one device.

4. The system of claim 2, wherein the digital data is accessed by a user connecting to the server through the communications network using a client software.
5. The system of claim 3, wherein the sensor data is used to derive ridership experience, road quality, and/or road roughness levels.
6. The system of claim 1, wherein image processing instructions include one or more neural networks for the purpose of processing the digital images and generating the object data and the resultant data.
7. The system of claim 6, wherein the system infers, using the neural networks, for purposes other than, or in addition to, generating the object data and the resultant data.
8. The system of claim 1, wherein some or all of the cameras and/or the sensors reside externally to the housing.
9. The system of claim 1, wherein the system includes a privacy and/or security framework.
10. The system of claim 1, wherein a device installation in the vehicle using the mounting component is permanent.
11. The system of claim 3, wherein the data collected by the at least one device includes the object data and the resultant data selected from the group consisting of: (a) cracks, including some or all of the following crack types: longitudinal crack(s), linear crack(s), transverse crack(s), reflection crack(s), lane joint crack(s), widening crack(s), fatigue crack(s), alligator crack(s), crocodile crack(s), block crack(s), hair crack(s), edge crack(s), edge joint crack(s), slippage crack(s), and/or shrinkage crack(s); (b) deformations and/or distortions in the pavement, including: rutting, shoving, corrugation, depression, and/or upheaval; (c) road repairs, including: asphalt patching, pothole patching and/or crack sealing; (d) road damage, including: potholes, stripping or ravelling; (e) street signage issues including: unreflective signs, bent signs, faded signs, obstructed signs, damaged signs, crooked sign, twisted sign, and/or vandalized sign; (f) manhole issues, including: such as raised manhole, sunken manhole, and/or open manhole; (g) drainage related issues, including: water pooling on pavement and/or clogged catch basins; (h) pavement marking issues, including: faded markings, and/or unreflective lane markings; (i) road obstruction issues, including: debris on road, cadavers, and/or construction garbage bins/dumpsters which are obstructing the right of way; and (j) side walk issues, including: trip edges, cracks, chips, distortion and/or deformation.
12. The system of claim 3, wherein the system derives road ratings, such as pavement condition index, for the roads which are travelled by the at least one device.
13. The system of claim 1, wherein the at least one device is used in the vehicle of a vehicle type selected from the group consisting of: (a) a car; (b) a truck; (c) a golf car; (d) an ATV; (e) a bicycle; (f) a motorbike; (g) a motorcycle; (h) a snowmobile; (i) a van; (j) a utility vehicle; (k) a garbage truck; (l) a sweeper; (m) a bus; (n) a school bus; (o) a van; (p) a taxi; and (q) an autonomous vehicle.
14. The system of claim 1, wherein the data is correlated with road segments and other assets to provide additional insights.
15. The system of claim 3, wherein the system derives road ratings, such as pavement condition index, for the roads which are covered by the at least one device.
16. The system of claim 1, wherein the data is collected from a private vehicle owned by an individual.
17. The system of claim 1, wherein the at least one device includes an integrated and/or external display.
18. The system of claim 17, wherein the display provides vehicle navigation functions.
19. The system of claim 1, wherein the at least one device runs one or more separate, third party applications, concurrently with the software instructions.
20. A method for collecting, processing, discarding, storing, transmitting, and disseminating digital

data, the data includes information captured from a device, a camera and a sensor mounted in a vehicle, and generated object data and generated resultant data.